AI-Powered Assistive Technologies for the Visually Impaired

Guide: Nagaraju Y
Assistant Professor, Department of Computer Science and Engineering
Dayananda Sagar College of Engineering (DSCE), Bengaluru, India
nagarajuyat22@gmail.com

Om Shivarjun T M
Department of Computer Science and Engineering
Dayananda Sagar College of Engineering (DSCE)
Bengaluru, India
omshivarjun2004@gmail.com

Sunder Bharti
Department of Computer Science and Engineering
Dayananda Sagar College of Engineering (DSCE)
Bengaluru, India
sunderbharti02@gmail.com

Shrilaxmi Pattar
Department of Computer Science and Engineering
Dayananda Sagar College of Engineering (DSCE)
Bengaluru, India
pattarshrilakshmi@gmail.com

Ajay Rathod
Department of Computer Science and Engineering
Dayananda Sagar College of Engineering (DSCE)
Bengaluru, India
arathod39869@gmail.com

***Abstract*—Assistive artificial intelligence for blind and visually impaired users is Assistive AI for blind and visually impaired users has advanced from isolated single-purpose tools narrating individual objects or reading isolated labels toward systems capable of sustaining a full conversational loop grounded in live camera input, spoken language, and personal memory. Yet most deployed solutions still depend on dedicated hardware, mobile applications, or cloud services that treat each interaction as stateless, leaving users without continuity across sessions and without protection against poor-quality visual input that degrades model reliability. This paper presents a voice and vision assistant system: a browser-native, hardware-free real-time assistant designed specifically for blind and visually impaired users, operable on any laptop with a webcam and microphone without application installation or specialised equipment. The system integrates a persistent bidirectional network transport layer, a dual-path vision architecture separating low-latency conversational responses from high-fidelity frame analysis, an eight-class language-model-based intent classifier that routes spoken queries to functionally specialised handlers, a heuristic capture quality coach that enforces frame quality thresholds before vision inference to prevent hallucination on degraded input, and a local semantic memory layer using 1024-dimensional text embeddings with cosine-similarity retrieval for cross-session personalisation. A vision-assisted web search route allows the system to identify physical objects through the camera before generating targeted search queries, extending assistance beyond static scene description into live information retrieval. An online self-correction mechanism logs user feedback signals to adapt response tone and verbosity dynamically across turns without retraining the underlying model. The system is evaluated across seven functional categories spanning scene description, optical character recognition, multilingual translation, object identification, memory recall, web search, and adaptive self-correction, encompassing thirty-eight structured test scenarios that include both successful inference paths and coach-blocked failure cases. Results demonstrate that disciplined turn handling, capture quality gating, and lightweight semantic memory together produce a practically deployable assistive loop that does not require dedicated infrastructure beyond a standard browser session.**

***Index Terms*—blind-first assistance, multimodal accessibility, voice interaction, vision-language systems, semantic memory, capture quality, WebSocket**

I. Introduction

The rapid expansion of multimodal large language models and real-time speech processing capabilities has opened a genuinely new design space for assistive technology — one where a blind or visually impaired user can hold a natural spoken conversation with a system that simultaneously sees, remembers, and searches on their behalf. For the approximately 285 million people worldwide living with visual impairment, daily tasks that sighted individuals perform effortlessly — reading a medicine label, identifying an unfamiliar object, translating a sign in a foreign language, or recalling a fact stored from a previous session — remain disproportionately dependent on either human assistance or narrowly scoped single-function tools that handle one task at a time and retain no memory of the user across interactions.

Early assistive systems addressed individual subtasks in isolation. Screen readers handled text, dedicated OCR applications extracted printed content, and remote sighted-assistance platforms connected blind users with volunteer helpers over video calls. While each approach demonstrated clear value within its intended scope, they share a structural limitation: each requires the user to consciously select the right tool for each situation, switch between applications, and reconstruct context manually across interactions. As generative models

matured and multimodal inference became practically viable at low latency, the research community began exploring whether a single conversational interface could subsume these fragmented capabilities into one coherent assistive loop driven entirely by natural speech.

Recent work on vision-language models and omni-modal architectures has brought that possibility closer to reality, but most published systems still evaluate in controlled laboratory conditions, rely on mobile hardware with dedicated neural processing units, or assume persistent high-bandwidth connections that are unavailable to many users. A further gap that the literature has been slow to address is the problem of degraded visual input. A blind user cannot verify whether the camera is pointed correctly, whether the lighting is adequate, or whether the page is tilted before submitting a frame for analysis. Systems that pass every captured frame to an inference model without quality checking produce hallucinated or meaningless responses that erode user trust, yet no lightweight, speech-integrated quality gating mechanism has been systematically deployed in a browser-native context. Persistent personalised memory across sessions represents a third gap: existing tools treat every interaction as independent, requiring users to re-establish preferences, correct recurring misunderstandings, and repeat contextual facts that the system should have retained.

This paper addresses these three gaps by presenting a voice and vision assistant system built entirely within the browser environment, requiring no dedicated application, no specialised hardware, and no manual tool selection by the user. The proposed system combines real-time spoken interaction, dual-path vision inference, LLM-based intent classification, heuristic frame quality gating, cross-session semantic memory, vision-assisted web search, and adaptive self-correction into a unified assistive loop accessible from any standard laptop.

A. Purpose

The purpose of this work is to design, implement, and evaluate a browser-native voice and vision assistant that integrates complementary capabilities — perception, speech, memory, and retrieval — into a single conversational interface for blind and visually impaired users, motivated by the fragmentation and deployment limitations identified in existing assistive technology

B. Motivation

Most existing assistive tools address one capability at a time and retain no state across user sessions, forcing users to manage tool selection and context reconstruction manually. Integrating real-time vision, spoken dialogue, intent routing, capture quality control, and persistent semantic memory into a single browser-native interface removes those coordination burdens and brings practical multimodal assistance within reach of any user with a standard laptop and internet connection.

C. Contributions

The main contributions of this paper are:

- A browser-native, hardware-free voice and vision assistant deployable on any standard laptop without application installation, integrating scene description, OCR, translation, object identification, memory, web search, and self-correction into a single spoken interface.
- A dual-path vision architecture that separates low-latency streaming conversational responses from high-fidelity direct-request frame analysis, routing between paths based on classified user intent.
- An eight-class language-model-based intent classifier that maps free-form spoken queries to functionally specialised handlers without requiring labelled training data from the target deployment domain.
- A heuristic capture quality coach that evaluates incoming frames for luminance sufficiency, local contrast distribution, and spatial uniformity before committing to vision inference, providing spoken corrective guidance when quality thresholds are not met and preventing hallucination on degraded input.
- A local semantic memory layer using 1024-dimensional text embeddings and cosine-similarity retrieval, supporting cross-session fact persistence, priority-fact preloading at session start, and similarity-aware deduplication without requiring external vector database infrastructure.
- A vision-assisted web search route that identifies physical objects through the camera before generating targeted search queries, extending the assistant beyond static description into live information retrieval.
- An online self-correction mechanism that logs user feedback signals and adapts response tone and verbosity dynamically across turns without modifying underlying model weights.

D. Research Questions

This work is guided by three research questions:

- **RQ1:** Can a browser-native voice and vision assistant coordinate multimodal inference, memory, and search without specialized hardware?
- **RQ2:** How does capture-quality gating affect reliability when users submit real camera frames under imperfect lighting, blur, or crop conditions?
- **RQ3:** How can local semantic memory reduce repeated explanations while preserving a simple deployment model?

II. Related Work

Assistive technology for blind and visually impaired users has been studied across accessibility, computer vision, natural language processing, and human-computer interaction. The proposed system builds on five streams of work: visual question answering from blind-user images, image description and personalization, remote assistance, retrieval and long-term memory, and multimodal model architectures.

A. Blind-User Visual Question Answering

VizWiz introduced an important interaction pattern for blind users: capture an image, ask a spoken visual question, and

receive an answer in near real time [2]. Later, the VizWiz visual question answering dataset formalized this setting and showed that blind-user photographs differ from standard vision datasets because images may be poorly framed, blurry, or unanswerable [3]. These findings are central to this paper because they show why capture quality and uncertainty handling are not peripheral details. A blind-first assistant must expect imperfect visual input and must decide when not to answer.

B. Image Description, Personalization, and Representation

Image descriptions for blind and low-vision users are not one-size-fits-all. Stangl et al. showed that users want different levels of detail depending on task and context [6]. Related work on image descriptions also highlights risks around privacy, identity, and representation, especially when descriptions concern people or socially sensitive content [5], [7]. These studies motivate short, task-aligned answers, uncertainty language, and correction logging rather than a single fixed verbosity style.

C. Remote Assistance and Teachable Recognition

Remote sighted assistance demonstrates that technical accuracy alone is not enough: trust, turn timing, repair language, and handoff quality strongly affect whether users can complete tasks [8]–[10]. Work on personalized object recognition and teachable AI further shows that users often care about recurring personal items, not only generic object categories [11], [12]. These findings support the use of persistent memory and user-controlled correction in the proposed system.

D. Retrieval and Long-Term Memory

Retrieval-augmented generation separates model parameters from externally retrieved facts, improving grounding and updateability for knowledge-intensive tasks [13]. Dense Passage Retrieval showed that learned dense representations can retrieve passages effectively for open-domain question answering [14], while REALM incorporated retrieval into language model pre-training [15]. MemoryBank and related long-term memory work show how conversational agents can store and reuse user-specific context over time [16]. The proposed system adopts a simpler local design: SQLite stores memories and embeddings, and cosine similarity retrieves relevant facts when needed.

E. Vision-Language and Multimodal Models

Modern multimodal systems build on the Transformer architecture [17], visual question answering [4], and vision-language pretraining. LXMERT, UNITER, and OSCAR showed how cross-modal pretraining can align visual and textual information for tasks such as VQA and image-text reasoning [18]–[20]. Vision Transformer work further demonstrated that attention-based architectures can operate directly on image patches [21]. Current commercial multimodal APIs extend these ideas into realtime speech, image, and video interfaces, making browser-based assistive loops feasible [28], [32], [33].

III. Proposed Methodology

The proposed methodology treats assistance as a loop rather than a one-shot prediction. A user speaks a request, the system captures or uses the current visual context, the router selects a task path, the capture coach decides whether the visual input is usable, the model path generates an answer, and the response is spoken back to the user. If the user corrects the answer, the correction is logged and influences later turns.

Figure 1 positions the proposed system relative to existing assistive categories. Voice-only assistants are useful for general questions but lack camera grounding. Camera-based reading tools can handle text but often lack dialogue, memory, or search. Remote assistance offers human guidance but requires handoff and raises privacy concerns. The proposed design combines these capabilities inside a browser-native system that can route between fast conversation and slower visual analysis.

Table I replaces a purely descriptive evaluation lens with a numerical rubric. The weights are project weights, not public benchmark scores. They reflect the fact that this system is judged not only by model accuracy but also by speech timing, frame repair, memory usefulness, and safe recovery from uncertain visual input.

IV. System Architecture and Implementation

The implementation follows a layered browser-backend-model structure. The browser handles microphone access, camera preview, user controls, and spoken output. The backend receives user turns over WebSocket, maintains session state, runs routing and capture checks, calls model services, stores memory and correction records, and returns text or audio responses.

Figure 2 shows the main pipeline. Browser input is sent to a WebSocket endpoint, which supports bidirectional communication without repeated polling. The WebSocket API is appropriate for a conversational assistive interface because it opens a two-way interactive communication session between the browser and the server [31]. The realtime speech path uses the Qwen-Omni realtime model family for low-latency audio interaction, while heavy vision tasks use a stronger visual reasoning path for OCR, object recognition, translation, and page-level understanding. Alibaba Cloud documents Qwen-Omni-Realtime as a realtime audio and video chat model that processes streaming audio and image inputs and returns text and audio output [28]. For memory, the system stores text embeddings and fact records locally in SQLite; Alibaba Cloud documents text embedding models as dense vectors for semantic search, recommendation, clustering, and related retrieval tasks [30].

A. Layered Components

The browser layer captures microphone input and camera frames. The transport layer carries audio turns and optional image frames through WebSocket. The orchestration layer classifies the user's spoken request and chooses whether to use realtime dialogue, heavy visual analysis, memory retrieval, web search, or correction handling. The evidence layer stores

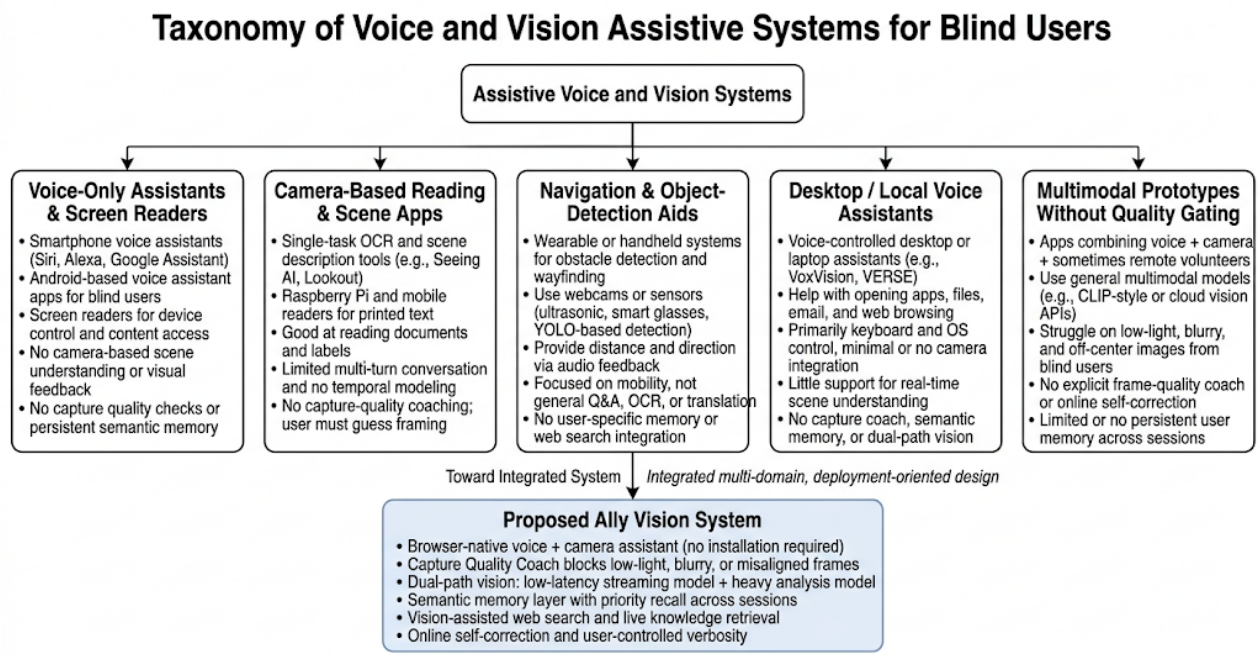

Fig. 1: Taxonomy of voice and vision assistive systems for blind users. The figure positions the proposed browser-native assistant against voice-only assistants, camera-based reading tools, navigation aids, desktop assistants, and multimodal prototypes that do not include capture-quality gating.

TABLE I: Weighted Evaluation Rubric for Blind-First Multimodal Assistance

Evaluation dimension	Weight (%)	Primary metric	Scoring range	Why it matters
Scene description and VQA	15	task success	0–100	Captures whether the answer helps the user understand a live scene.
OCR/document reading	20	CER/WER accuracy	0–100	Reading text is a core use case for documents, labels, receipts, and forms.
Realtime speech interaction	15	response delay	0–100	Slow speech response makes the assistant harder to use in everyday interaction.
Capture-quality recovery	15	rejection accuracy	0–100	A blind user may not know whether the frame is usable, so repair guidance is essential.
Memory and personalization	10	recall precision	0–100	Persistent recall reduces repeated explanations across sessions.
Web search and grounding	10	answer support	0–100	Current facts should be retrieved rather than guessed.
Self-correction logging	15	correction success	0–100	Repeated errors should change later behavior instead of being ignored.
Total	**100**	–	–	–

recent frames, transcripts, memory records, embeddings, and correction logs. This division keeps the interface simple while allowing internal specialization.

B. Capture Quality Gate

The capture coach is placed before heavy visual inference. It checks for low luminance, insufficient contrast or sharpness, excessive uniformity, and poor usable visual area. When the frame fails these checks, the system returns a spoken repair instruction such as asking the user to move closer, increase light, or reposition the page. This avoids sending obviously weak frames to a vision model and reduces the risk of confident hallucination.

V. Numerical Component Selection and Comparison

Component selection was based on numerical evidence where public benchmark values are available and on clearly labeled project criteria where no public benchmark exists. MMMU is used as a reference for multimodal reasoning, MathVista is used for visual mathematical reasoning, and OCRBench v2 is used for text-centric image understanding [22]–[24]. These benchmarks are useful, but they do not fully measure blind-first interaction. A model with a high benchmark score can still be a poor deployment choice if it has high latency, cross-provider complexity, or weak integration with the rest of the system. For that reason, the tables below combine public values, N/A markers for unavailable numbers, and a project suitability score calculated with the scoring method in Section VI.

Table II shows why OCR and document metrics are more

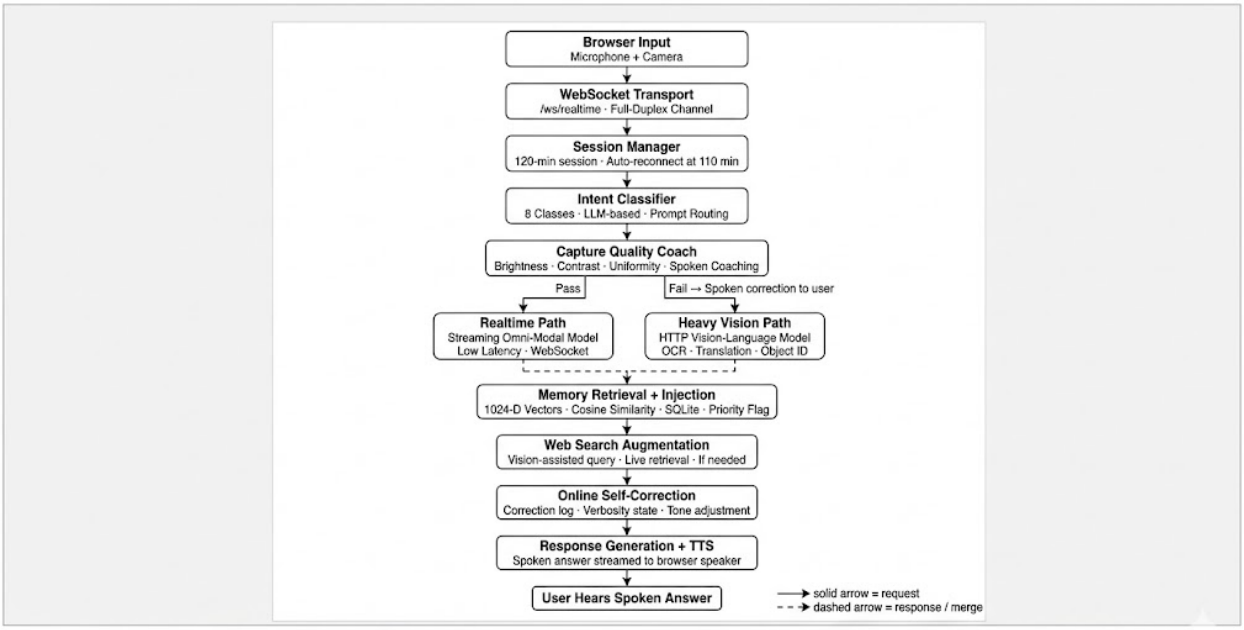

Fig. X. Overall system architecture of the proposed voice and vision assistant for blind users showing browser-side capture, intent routing, dual-path vision inference, semantic memory, web search augmentation, self-correction, and TTS spoken output.

Fig. 2: Overall system architecture of the proposed voice and vision assistant system, including browser capture, WebSocket transport, intent routing, capture quality gating, dual-path inference, memory retrieval, web search augmentation, self-correction, and spoken output.

TABLE II: Numerical Heavy Vision and Multimodal Model Comparison

Model or public reference	MMMU (%)	MathVista (%)	OCRBench v2	Context (K tokens)	Cost index (1–5)	Project score /100
Qwen3.6 Plus	86.0	N/A	N/A	1000	4	92
Qwen3.5 Flash	N/A	N/A	N/A	N/A	2	78
Qwen2.5-VL-72B reference	70.2	N/A	63.7	N/A	3	84
GPT-4o	72.2	63.8	N/A	128	5	83
Gemini 2.0 Flash	70.7	N/A	N/A	1000	2	80
Pixtral Large reference	64.0	N/A	N/A	128	3	70

relevant to this project than generic captioning alone. The Qwen heavy-vision path receives a high project score because it aligns with the selected DashScope stack and avoids cross-provider orchestration. GPT-4o and Gemini Flash remain important reference systems, but their use would add provider switching and additional integration rules. N/A values are retained instead of being filled with guesses, because not every deployed endpoint has a public score on every benchmark.

Table III is intentionally conservative about latency. Official public documentation often describes realtime behavior but does not provide a stable end-to-end latency number for every environment. The session-limit and modality columns are therefore more reliable for architectural comparison. The Qwen realtime path receives the highest project score because it matches the backend's DashScope route, supports audio plus image interaction, and avoids stitching separate ASR, LLM, and TTS services.

Table IV favors SQLite plus embeddings because the intended scale is a student prototype and a single-user or small-user memory store. Alibaba Cloud lists text-embedding-v4 and text-embedding-v3 with selectable dimensions and 8,192-token single-line input limits, which is enough for short memory facts and summaries [30]. Vector databases and FAISS-style indexes are useful at larger scale, but they add deployment components that are not necessary for the present system.

Table V uses project evaluation targets rather than public benchmark scores because capture coaching is an interaction mechanism, not a standard foundation-model benchmark. The numerical targets make the evaluation plan concrete: the capture coach should reject most unusable frames with low latency, while the correction log should improve later behavior even though it is not itself a frame filter.

VI. Scoring and Equations

The formulas below describe routing, suitability scoring, dual-path inference, capture gating, semantic memory retrieval, and self-correction. Each equation is kept within a single IEEE column.

TABLE III: Numerical Realtime Speech and Omni Model Comparison

Model or option	Latency (ms)	Input modalities	Output modalities	Session limit (min)	Interrupt support (0/1)	Project score /100
qwen3.5-omni-plus-realtime	N/A	3	2	120	1	94
qwen3.5-omni-flash-realtime	N/A	3	2	120	1	86
GPT-4o Realtime	N/A	2	2	60	1	82
Gemini Live	N/A	3	2	N/A	1	81
STT–LLM–TTS pipeline	N/A	1	1	N/A	0	62

TABLE IV: Numerical Embedding and Memory Architecture Comparison

Option	Embedding dim.	Max input tokens	Local persistence (0/1)	Extra service (0/1)	Complexity /10	Project score /100
SQLite + text-embedding-v4	1024	8192	1	1	2	95
SQLite + text-embedding-v3	1024	8192	1	1	2	92
Chroma local store	N/A	N/A	1	0	5	72
Qdrant deployment	N/A	N/A	1	1	6	65
Pinecone hosted vector DB	N/A	N/A	0	1	7	55
FAISS local index	N/A	N/A	1	0	5	68

TABLE V: Numerical Capture Coach and Self-Correction Evaluation Targets

Option	Poor-frame rejection (%)	Added latency (ms)	Risk reduction /100	Spoken repair (0/1)	Logging (0/1)	Project score /100
Capture Coach	85	40	80	1	0	90
No quality filter	0	0	0	0	0	20
OpenCV blur/brightness gate	70	25	65	1	0	76
Model-based frame judge	90	1200	88	1	0	78
Custom correction/reflection log	0	20	35	0	1	82
No correction	0	0	0	0	0	15

A. Normalized Suitability Score

The project score used in Tables II–V is computed from normalized criteria:

$$S_j = 100 \sum_{k=1}^{K} w_k \tilde{x}_{j,k}, \qquad \sum_{k=1}^{K} w_k = 1. \tag{1}$$

Here S_j is the final suitability score for candidate j, w_k is the weight of metric k, and $\tilde{x}_{j,k}$ is a normalized value between 0 and 1. Higher-is-better metrics are normalized as $x/\max(x)$, while lower-is-better metrics such as latency or deployment complexity use $\min(x)/x$. Missing values are marked N/A and excluded from the public-benchmark part of the score.

B. Routing

For each spoken turn, the router selects a functional route from the route set:

$$\hat{r} = \arg\max_{r \in \mathcal{R}} S(r \mid x_s, x_v, m, h). \tag{2}$$

Here x_s is the spoken transcript, x_v is visual input, m is retrieved memory context, h is recent session history, and $\mathcal{R}$ includes scene description, OCR, translation, object recognition, memory recall, memory write, web search, and correction routes.

C. Dual-Path Inference

The system separates fast conversation from detailed visual reasoning:

$$P(y|x_s, x_v) = \begin{cases} P_{rt}(y|x_s, x_v, h), & r = \mathrm{RT}, \\ P_{hv}(y|x_s, x_v, p), & r = \mathrm{HV}. \end{cases} \tag{3}$$

The realtime path P_{rt} is used when a short spoken answer is more important than detailed visual analysis. The heavy-vision path P_{hv} is used when the task involves OCR, document understanding, translation, or careful visual comparison.

D. Capture Quality Gate

Before heavy visual inference, the capture coach applies a binary gate:

$$Q = \mathbf{1}[L \geq \tau_L] \wedge \mathbf{1}[C \geq \tau_C] \wedge \mathbf{1}[U \leq \tau_U] \wedge \mathbf{1}[A \geq \tau_A]. \tag{4}$$

Here L is luminance, C is contrast or sharpness, U is uniformity, and A is usable visual area. If $Q = 0$, the assistant gives spoken repair guidance instead of returning a visual answer.

E. Semantic Memory Retrieval

Memory retrieval uses cosine similarity between a query embedding and stored memory embeddings:

$$\mathrm{sim}(\mathbf{q}, \mathbf{k}_i) = \frac{\mathbf{q} \cdot \mathbf{k}_i}{\|\mathbf{q}\| \, \|\mathbf{k}_i\|}. \tag{5}$$

A stored memory is used only when its similarity is high enough and its category is appropriate for the current task.

F. Self-Correction Caution Update

When the user signals that an answer is wrong, too long, too short, or not useful, the system increases a session-level caution state:

$$c_t = \min(1.0, c_{t-1} + \alpha I_t). \tag{6}$$

The indicator I_t is 1 when a correction signal is detected and 0 otherwise. Higher caution makes later answers more conservative and more explicit about uncertainty.

VII. Functional Demonstration

The implementation was demonstrated through camera-grounded tasks. The screenshots are not used as formal benchmark proof; they provide qualitative evidence that the interface, routing, visual grounding, and spoken response flow operate in the intended manner.

The object-recognition route demonstrates the heavy-vision path for identifying physical objects. The OCR route demonstrates document or page reading from camera input. Figure 5 is used for the remaining functions because the visual layout remains the same: live camera on one side, conversation history on the other, and spoken interaction as the primary user interface.

VIII. Evaluation Design

The evaluation design is organized by task family rather than by model alone. Blind-first assistance is an interaction problem: the assistant must capture useful input, choose the right route, answer concisely, recover from weak frames, and avoid false memory. The planned test set contains 38 structured scenarios across seven groups:

- scene description: desk scene, dark room, movement, person in frame, and multiple objects;
- OCR and reading: printed article, receipt, medicine label, handwriting, mathematical expression, and low-quality capture;
- multilingual translation: visible text in multiple languages with spoken output;
- object recognition: household object, product-like object, and mixed table scene;
- memory: save fact, recall fact, update fact, and reject unknown fact;
- web search: spoken query and vision-assisted query;
- self-correction: wrong answer, verbosity change, repeated correction, and caution update.

The evaluation metrics are numerical wherever practical: task completion rate, response latency, capture rejection accuracy, object and text coverage, memory recall precision, false-memory rejection, character error rate, word error rate, and correction success. Human relevance ratings are still needed for scene description because object-count accuracy alone does not measure whether a spoken description is useful to a blind user.

IX. Discussion

The numerical tables change the interpretation of component selection. A high MMMU or MathVista score indicates strong multimodal reasoning, but it does not directly measure spoken repair, camera framing support, or memory continuity. For this project, OCR/document behavior and capture recovery are more important than general image-captioning performance because the assistant is expected to read pages, labels, and visible text under imperfect laptop-camera conditions.

The results also explain why the same-vendor DashScope path is preferred even when other models have strong benchmark results. A single provider for realtime speech, heavy vision, embeddings, and optional search reduces integration overhead. This does not make other models weak; it means the deployment problem is broader than model accuracy. In a student prototype intended to run from a browser and a FastAPI backend, simpler integration is an important part of reliability.

Local persistence is another practical choice. SQLite with embeddings is not an industrial-scale vector database, but it is inspectable, portable, and sufficient for user facts, short summaries, and correction records. For a single-user or small-user assistive prototype, this is a better fit than adding an external vector database before the need for scale has been demonstrated.

X. Limitations and Future Work

The current system remains a prototype. First, capture-quality thresholds may not generalize across all webcams, rooms, paper types, and lighting conditions. Second, some table values are marked N/A because public benchmark numbers are not available for every exact deployed endpoint. Third, prompt-based routing can fail on compound requests, such as asking the assistant to read a label and search for the product in the same turn. Fourth, semantic memory needs user-facing controls for editing, deleting, and prioritizing stored facts. Fifth, self-correction currently depends mainly on explicit feedback; detecting implicit dissatisfaction remains difficult. Sixth, the evaluation scenarios should be extended into studies with blind and visually impaired participants before making strong claims about real-world accessibility impact.

Future work should add adaptive threshold calibration, stronger document-session question answering, editable memory controls, richer uncertainty handling, and formal usability studies. A future version should also compare the dual-path design against a single-model baseline to measure how much routing and capture gating improve practical task success.

XI. Conclusion

This paper presented AI-Powered Assistive Technologies for the Visually Impaired, a browser-native voice and vision assistant for blind and visually impaired users. The system combines realtime conversation, heavy visual analysis, capture-quality gating, local semantic memory, web search, and self-correction in one interaction loop. The numerical comparison shows that benchmark strength matters, but it is only one part of the design. A practical assistant also needs fast speech

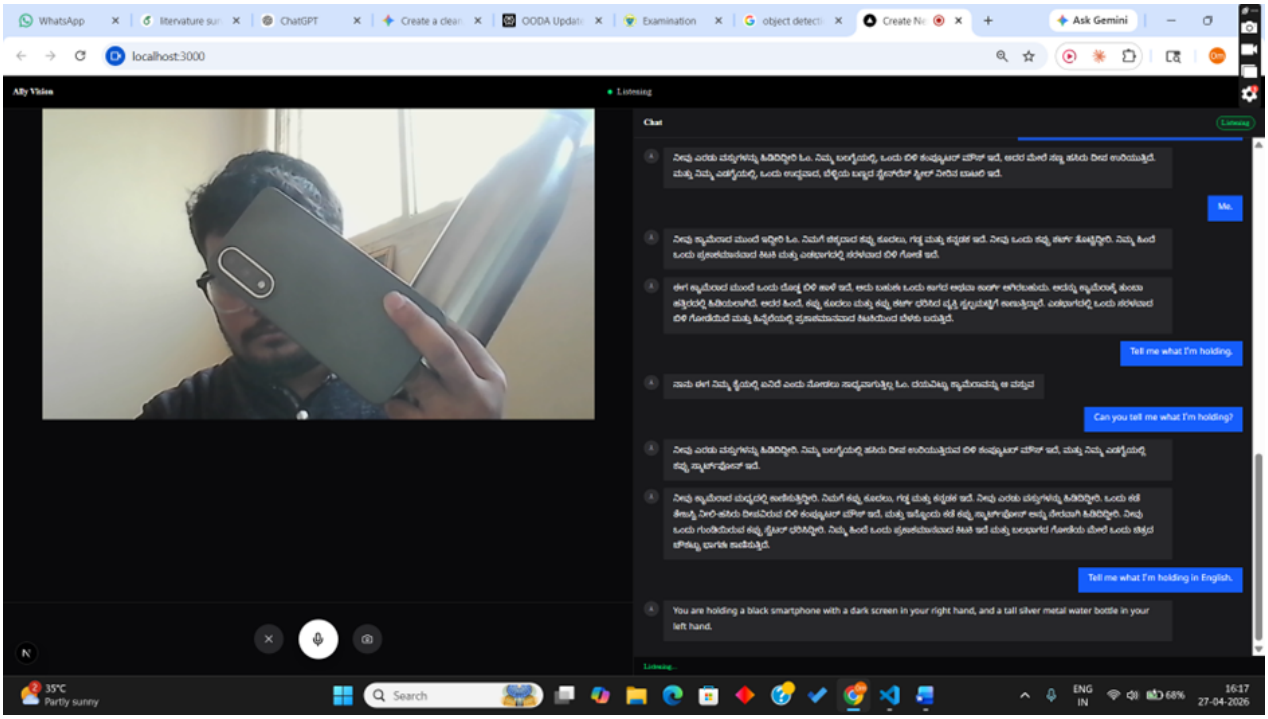

Fig. 3: Object-recognition demonstration using the project interface. The user presents an object to the camera and asks for identification.

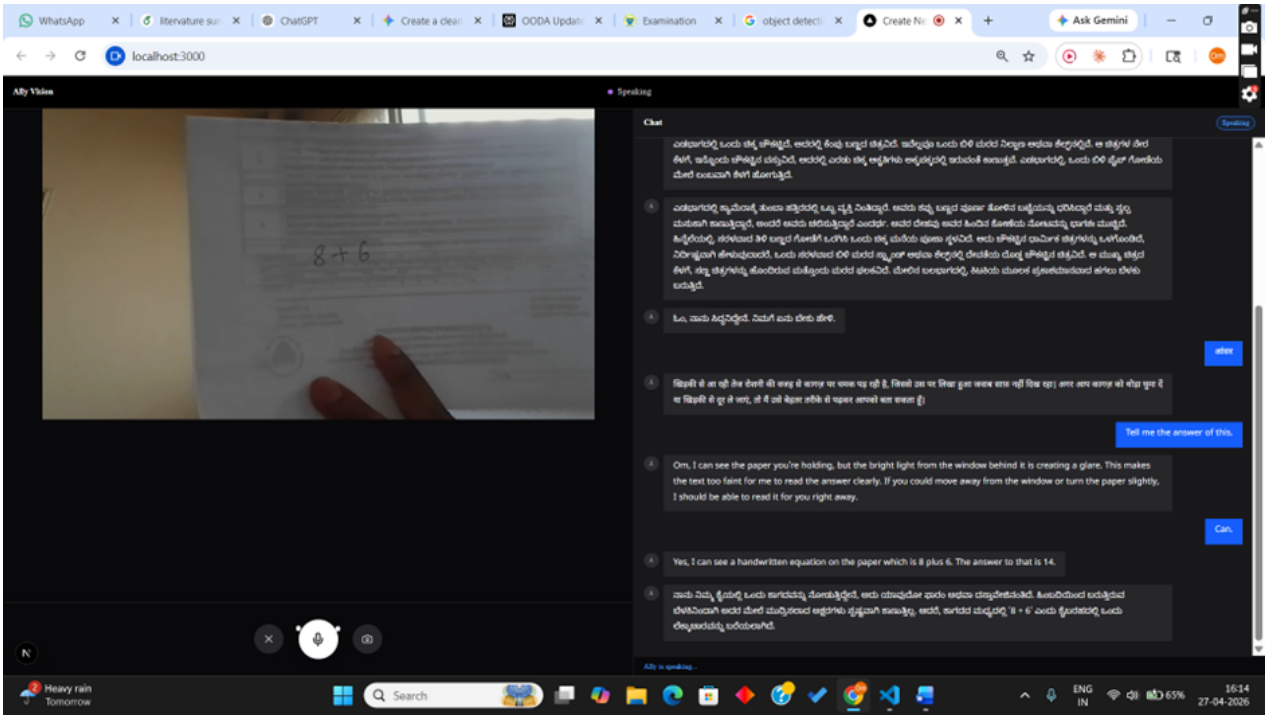

Fig. 4: OCR/text-reading demonstration using camera-grounded visual input. The assistant reads visible text when the frame is sufficiently usable and gives repair guidance when it is not.

interaction, capture repair, simple deployment, and persistent memory. Rather than treating a foundation model as the complete solution, the system shows how routing, repair, and memory can make multimodal assistance more usable in everyday conditions.

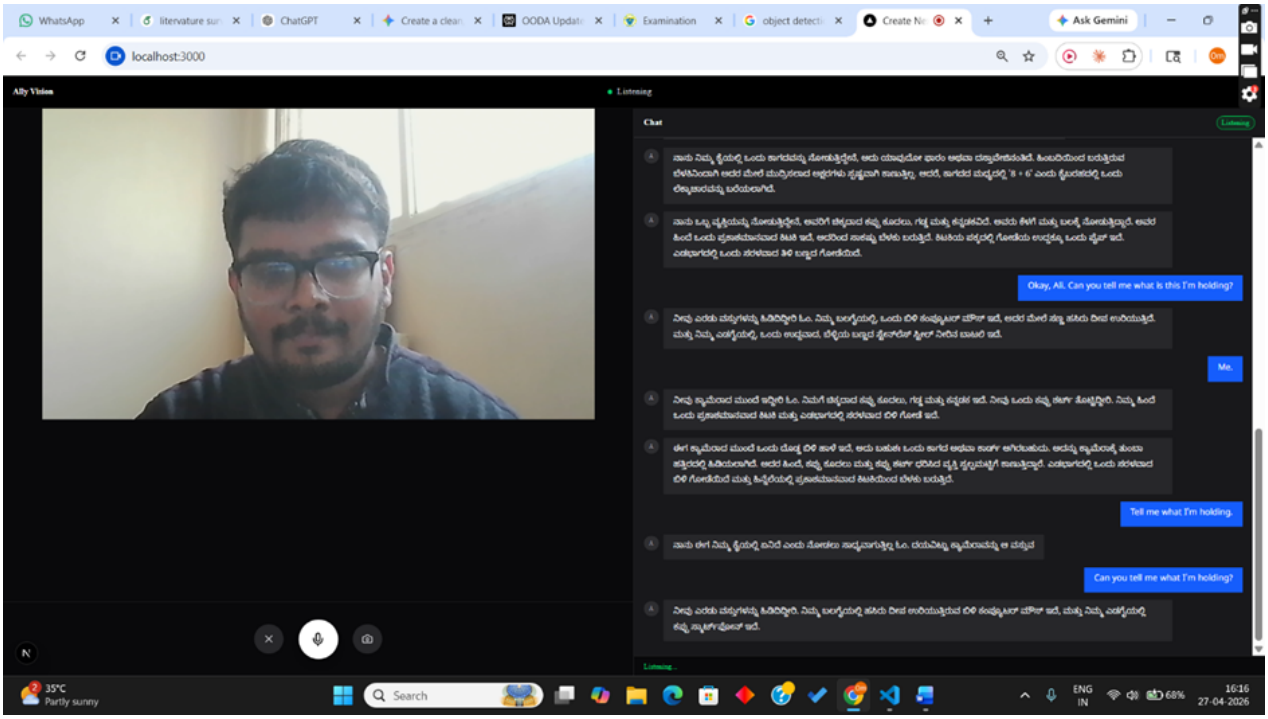

Fig. 5: Representative interface output for the remaining implemented functions. This screenshot represents scene description, multilingual response, memory recall, self-correction, and web search, where the interface layout remains mostly the same.

REFERENCES

[1] World Health Organization, "Blindness and vision impairment," WHO Fact Sheet, 2026.

[2] J. P. Bigham, C. Jayant, H. Ji, G. Little, A. Miller, R. C. Miller, R. Miller, A. Tatarowicz, B. White, S. White, and T. Yeh, "VizWiz: Nearly real-time answers to visual questions," in *Proc. ACM Symposium on User Interface Software and Technology*, 2010, pp. 333–342.

[3] D. Gurari, Q. Li, A. J. Stangl, A. Guo, C. Lin, K. Grauman, J. Luo, and J. P. Bigham, "VizWiz grand challenge: Answering visual questions from blind people," in *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2018, pp. 3608–3617.

[4] S. Antol, A. Agrawal, J. Lu, M. Mitchell, D. Batra, C. L. Zitnick, and D. Parikh, "VQA: Visual question answering," in *Proc. IEEE International Conference on Computer Vision*, 2015, pp. 2425–2433.

[5] A. Stangl, M. R. Morris, and D. Gurari, "Person, shoes, tree. Is the person naked? What people with vision impairments want in image descriptions," in *Proc. CHI Conference on Human Factors in Computing Systems*, 2020.

[6] A. Stangl, N. Verma, K. R. Fleischmann, M. R. Morris, and D. Gurari, "Going beyond one-size-fits-all image descriptions to satisfy the information wants of people who are blind or have low vision," in *Proc. ACM SIGACCESS Conference on Computers and Accessibility*, 2021.

[7] C. L. Bennett, C. Gleason, M. K. Scheuerman, J. P. Bigham, A. Guo, and A. To, "It is complicated: Negotiating accessibility and misrepresentation in image descriptions of race, gender, and disability," in *Proc. CHI Conference on Human Factors in Computing Systems*, 2021.

[8] S. Lee, M. Reddie, C.-H. Tsai, J. Beck, M. B. Rosson, and J. M. Carroll, "The emerging professional practice of remote sighted assistance for people with visual impairments," in *Proc. CHI Conference on Human Factors in Computing Systems*, 2020.

[9] J. Xie, R. Yu, K. Cui, S. Lee, J. M. Carroll, and S. M. Billah, "Are two heads better than one? Investigating remote sighted assistance with paired volunteers," in *Proc. Designing Interactive Systems Conference*, 2023.

[10] R. Yu, S. Lee, J. Xie, S. M. Billah, and J. M. Carroll, "Human-AI collaboration for remote sighted assistance: Perspectives from the LLM era," *Future Internet*, vol. 16, no. 7, 2024.

[11] D. Ahmetovic, D. Sato, U. Oh, T. Ishihara, K. M. Kitani, and C. Asakawa, "ReCog: Supporting blind people in recognizing personal objects," in *Proc. CHI Conference on Human Factors in Computing Systems*, 2020.

[12] C. Morrison, M. Grayson, R. F. Marques, D. Massiceti, C. Longden, L. Wen, and E. Cutrell, "Understanding personalized accessibility through teachable AI: Designing and evaluating Find My Things for people who are blind or low vision," in *Proc. ACM SIGACCESS Conference on Computers and Accessibility*, 2023.

[13] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Kuettler, M. Lewis, W. Yih, T. Rocktaschel, S. Riedel, and D. Kiela, "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Advances in Neural Information Processing Systems*, vol. 33, 2020.

[14] V. Karpukhin, B. Oguz, S. Min, P. Lewis, L. Wu, S. Edunov, D. Chen, and W. Yih, "Dense passage retrieval for open-domain question answering," in *Proc. Conference on Empirical Methods in Natural Language Processing*, 2020, pp. 6769–6781.

[15] K. Guu, K. Lee, Z. Tung, P. Pasupat, and M.-W. Chang, "REALM: Retrieval-augmented language model pre-training," in *Proc. International Conference on Machine Learning*, 2020.

[16] W. Zhong, L. Guo, Q. Gao, H. Ye, and Y. Wang, "MemoryBank: Enhancing large language models with long-term memory," in *Proc. AAAI Conference on Artificial Intelligence*, 2024.

[17] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, "Attention is all you need," in *Advances in Neural Information Processing Systems*, vol. 30, 2017.

[18] H. Tan and M. Bansal, "LXMERT: Learning cross-modality encoder representations from transformers," in *Proc. Conference on Empirical Methods in Natural Language Processing*, 2019.

[19] Y.-C. Chen, L. Li, L. Yu, A. El Kholy, F. Ahmed, Z. Gan, Y. Cheng, and J. Liu, "UNITER: Universal image-text representation learning," in *Proc. European Conference on Computer Vision*, 2020.

[20] X. Li, X. Yin, C. Li, P. Zhang, X. Hu, L. Zhang, L. Wang, H. Hu, L. Dong, F. Wei, Y. Choi, and J. Gao, "Oscar: Object-semantics aligned pre-training for vision-language tasks," in *Proc. European Conference on Computer Vision*, 2020.

[21] A. Dosovitskiy et al., "An image is worth 16x16 words: Transformers for image recognition at scale," in *Proc. International Conference on Learning Representations*, 2021.

[22] X. Yue et al., "MMMU: A massive multi-discipline multimodal understanding and reasoning benchmark for expert AGI," in *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024.
[23] P. Lu et al., "MathVista: Evaluating mathematical reasoning of foundation models in visual contexts," in *Proc. International Conference on Learning Representations*, 2024.
[24] H. Fu et al., "OCRBench v2: An improved benchmark for evaluating large multimodal models on visual text localization and reasoning," 2025.
[25] LLM-Stats, "MMMU leaderboard and model benchmark aggregation," 2026.
[26] LLM-Stats, "MathVista leaderboard and model benchmark aggregation," 2026.
[27] CodeSOTA, "OCRBench v2 leaderboard," 2026.
[28] Alibaba Cloud, "Qwen-Omni-Realtime," Model Studio Documentation, 2026.
[29] Qwen Team, "Qwen3.5-Omni: Scaling Up, Toward Native Omni-Modal AGI," Technical Blog, 2026.
[30] Alibaba Cloud, "Model Studio text embedding models and synchronous embedding API," Model Studio Documentation, 2026.
[31] MDN Web Docs, "WebSocket API," Mozilla Developer Network, 2026.
[32] OpenAI, "Realtime conversations and GPT-4o realtime model documentation," Technical Documentation, 2026.
[33] Google, "Gemini Live API documentation," Google AI for Developers, 2026.
[34] LLaVA Project, "LLaVA model cards and technical documentation," Hugging Face Model Documentation, 2024.
[35] OpenGVLab, "InternVL model cards and technical documentation," Hugging Face Model Documentation, 2024.
[36] Microsoft, "Phi vision model cards and technical documentation," Hugging Face Model Documentation, 2024.
[37] Mistral AI, "Pixtral model documentation," Technical Documentation, 2024.
[38] M. Shell, "IEEEtran: Document class for IEEE Transactions journals and conferences," CTAN Package Documentation, 2025.